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## Project Initialization and Planning Phase

**Date:** 15 March 2026

**Team ID:** *(Enter Your Team ID)*

**Project Title:** Human Development Index (HDI) Prediction Using Machine Learning

**Maximum Marks:** 5 Marks

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## Project Proposal (Proposed Solution) Report

The proposed project aims to develop an intelligent **Human Development Index (HDI) Prediction System** using Machine Learning. The system predicts the HDI category of a country by analyzing four important socio-economic indicators: **Life Expectancy, Mean Years of Schooling, Expected Years of Schooling, and Gross National Income (GNI) per Capita.**

The application is developed using **Python, Scikit-learn, Flask, HTML, CSS, and VS Code.** It provides a simple web interface where users can enter development indicators and instantly receive the predicted HDI category. The solution reduces manual analysis, improves prediction accuracy, and supports governments, researchers, students, and policymakers in understanding human development.

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## Project Overview

### Objective :-

The primary objective of this project is to build a Machine Learning model capable of predicting the Human Development Index category accurately using socio-economic indicators. The project also aims to provide a user-friendly web application for quick and reliable predictions.

### Scope: -

The project focuses on:

- Predicting the Human Development Index category.
- Analyzing development indicators.
- Supporting research and policymaking.

- Providing a Flask-based web application.
  - Improving decision-making using machine learning techniques.
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## **Problem Statement**

### **Description :-**

Determining the Human Development Index requires analyzing multiple socio-economic indicators simultaneously. Manual analysis is time-consuming, requires domain expertise, and may result in inaccurate conclusions. Governments and researchers need a faster, more reliable system for evaluating development levels.

### **Impact :-**

The proposed solution improves prediction accuracy, reduces manual effort, saves time, and enables better planning for education, healthcare, and economic development. It helps identify countries requiring development intervention and supports evidence-based policy decisions.

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### **Proposed Solution :-**

#### **Approach**

The project uses a **Machine Learning classification algorithm** trained on historical HDI data. Users enter the following indicators:

- Life Expectancy
- Mean Years of Schooling
- Expected Years of Schooling
- GNI Per Capita

The trained model processes these inputs and predicts one of the following HDI categories:

- Very High HDI
- High HDI
- Medium HDI
- Low HDI

The prediction is displayed through a Flask web application running in **Visual Studio Code (VS Code)**.

### Key Features

- Machine Learning-based HDI prediction.
- Fast and accurate prediction results.
- User-friendly Flask web interface.
- Data preprocessing and validation.
- Visualization using Matplotlib and Seaborn.
- Trained model stored using Joblib (model.pkl).
- Easy deployment and maintenance.

### Resource Requirements

#### Hardware

| Resource Type | Description         | Specification / Allocation                   |
|---------------|---------------------|----------------------------------------------|
| Processor     | Computing Resources | Intel Core i3 or above                       |
| Memory        | RAM                 | Minimum 4 GB                                 |
| Storage       | Disk Space          | Minimum 10 GB Free Space                     |
| Internet      | Network             | Required for package installation and GitHub |

#### Software

| Resource Type        | Description | Specification / Allocation |
|----------------------|-------------|----------------------------|
| Operating System     | Platform    | Windows / Linux / macOS    |
| Programming Language | Development | Python 3.x                 |
| Framework            | Backend     | Flask                      |

| Resource Type           | Description                      | Specification / Allocation                               |
|-------------------------|----------------------------------|----------------------------------------------------------|
| Libraries               | Machine Learning & Data Analysis | Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, Joblib |
| Development Environment | IDE                              | <b>Visual Studio Code (VS Code)</b>                      |
| Version Control         | Repository                       | Git & GitHub                                             |

## Data

| Resource Type | Description | Specification                                                                                                               |
|---------------|-------------|-----------------------------------------------------------------------------------------------------------------------------|
| Dataset       | HDI Dataset | CSV file containing Life Expectancy, Mean Years of Schooling, Expected Years of Schooling, GNI Per Capita, and HDI Category |

## Team Members

| Name                      | Role      | Responsibility                                                                           |
|---------------------------|-----------|------------------------------------------------------------------------------------------|
| <b>G Naveen</b>           | Team Lead | Project planning, GitHub management, model integration                                   |
| <b>Chandana Busireddy</b> | Member    | Dataset collection and preprocessing                                                     |
| <b>Gandham Joshna</b>     | Member    | Data visualization and exploratory data analysis                                         |
| <b>Nandineni Gowthami</b> | Member    | Flask web application development and testing                                            |
| <b>Shaik Talha</b>        | Member    | Machine learning model training, documentation, GitHub updates, and project presentation |

**Expected Outcome :-**

The system will successfully predict the **Human Development Index (HDI)** category using four key socio-economic indicators. Developed in **VS Code** with **Python and Flask**, it provides a practical, accurate, and user-friendly solution for governments, researchers, students, and policymakers to analyze and compare human development levels. Before submitting, replace the placeholder **Team ID** with your actual team number.